Subject: Prospective PhD Applicant (2027) | ZJU B.Eng. | AI for Medical Image Analysis | PhD-EAS Program

Dear Prof. Li,

My name is Shenshi Li, and I am currently a third-year undergraduate student in **Optoelectronic Information Science and Engineering at Zhejiang University.** I am writing to express my strong interest in pursuing a PhD under your supervision through the HKUST ECE PhD Early Admissions Scheme (**PhD-EAS**) for the **2027–28** intake.

Driven by my **family's medical background** and my father's work as a surgeon, **I have developed a long-standing commitment to bridging the gap between imaging technology and clinical needs**. Growing up, I was frequently exposed to medical knowledge and clinical stories, which made me deeply aware of the importance of medical imaging devices and accurate image-based diagnosis. After entering university, I actively reached out to my father's hospital to conduct informal visits and discussions, hoping that my technical training could one day contribute to real clinical needs. However, I also realized that my current knowledge is still at a relatively fundamental level, and that my home department has limited undergraduate projects related to medical imaging.

These experiences have motivated me to seek systematic training in AI for medical image analysis, and after learning about your research, I developed a strong desire to pursue my PhD in your group.

My academic performance ranks me **18/103 (GPA 3.94/4.0)**, supported by the **National Scholarship**. These honors ensure I am a strong candidate for my college's nomination to the PhD-EAS program.

My research focuses on the **intersection of optical imaging and deep learning-based analysis**, equipping me with a full-stack capability from hardware constraints to algorithm development:

- DL-based Defect Detection & DFF Reconstruction: As project leader, I developed a neural network-augmented pipeline to detect subtle surface defects in heterogeneous materials, handling the entire workflow from project planning to model deployment.
- Precision Optical Measurement: I designed OpenCV-based software for high-aspect-ratio micro-hole recognition. This experience sharpened my skills in robust image processing and handling complex optical data.

These experiences have solidified my belief that combining physical imaging principles with advanced AI—a core theme in your research—is the key to robust medical diagnosis. I am particularly inspired by your work on label-efficient and generalizable DL for CT/MRI. I am confident that my background in optical system design and practical AI implementation will allow me to contribute effectively to your group's research on clinically applicable AI.

I have attached my CV for your review. Would you be available for a brief meeting or email discussion to see if my background aligns with your recruitment plans for the 2027 intake? I would also value any advice on specific areas I should strengthen before the formal application.

Thank you for your time and consideration.

Best regards,

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